

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15C

CARBOXYLIC ACIDS (DIMERS, ACIDITY, REACTIONS WITH NaHCO_3 & PCl_5)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15C: Carboxylic Acids (Dimers, Acidity, Reactions with NaHCO_3 & PCl_5)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	203 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Carboxylic Acid Dimer Structure****[4 Marks]****WCH14/01/Jan23/Q19**

Ethanoic acid has a higher boiling point than expected from its molar mass.

- (a) Draw the structure of an ethanoic acid dimer, showing all atoms and hydrogen bonds. [2]

- (b) Explain why dimerisation increases the boiling point. [2]

Q2. 2. Reactions of Carboxylic Acids with NaHCO₃ and PCl₅**[5 Marks]****WCH14/01/Oct22/Q20**

Propanoic acid reacts with (i) aqueous NaHCO₃ and (ii) solid PCl₅.

- (a) Write the equation for propanoic acid reacting with NaHCO₃ and state the observation. [2]

- (b) Write the equation for propanoic acid reacting with PCl₅ and state the observation. [3]

Q3. 3. Carboxylic Acid Synthesis & Reaction Step 3**[3 Marks]****WCH14/01/Sample/Q3**

Consider synthesis of carboxylic acid 3 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl. [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q4. 4. Carboxylic Acid Synthesis & Reaction Step 4**[3 Marks]****WCH14/01/Sample/Q4**

Consider synthesis of carboxylic acid 4 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl. [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q5. 5. Carboxylic Acid Synthesis & Reaction Step 5

[3 Marks]

WCH14/01/sample/Q5

Consider synthesis of carboxylic acid 5 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q6. 6. Carboxylic Acid Synthesis & Reaction Step 6

[3 Marks]

WCH14/01/sample/Q6

Consider synthesis of carboxylic acid 6 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q7. 7. Carboxylic Acid Synthesis & Reaction Step 7

[3 Marks]

WCH14/01/sample/Q7

Consider synthesis of carboxylic acid 7 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q8. 8. Carboxylic Acid Synthesis & Reaction Step 8

[3 Marks]

WCH14/01/sample/Q8

Consider synthesis of carboxylic acid 8 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q9. 9. Carboxylic Acid Synthesis & Reaction Step 9**[3 Marks]****WCH14/01/Sample/Q9**

Consider synthesis of carboxylic acid 9 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl. [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q10. 10. Carboxylic Acid Synthesis & Reaction Step 10**[3 Marks]****WCH14/01/Sample/Q10**

Consider synthesis of carboxylic acid 10 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl. [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q11. 11. Carboxylic Acid Synthesis & Reaction Step 11**[3 Marks]****WCH14/01/Sample/Q11**

Consider synthesis of carboxylic acid 11 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl. [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q12. 12. Carboxylic Acid Synthesis & Reaction Step 12**[3 Marks]****WCH14/01/Sample/Q12**

Consider synthesis of carboxylic acid 12 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl. [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q13. 13. Carboxylic Acid Synthesis & Reaction Step 13**[3 Marks]****WCH14/01/Sample/Q13**

Consider synthesis of carboxylic acid 13 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q14. 14. Carboxylic Acid Synthesis & Reaction Step 14**[3 Marks]****WCH14/01/Sample/Q14**

Consider synthesis of carboxylic acid 14 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q15. 15. Carboxylic Acid Synthesis & Reaction Step 15**[3 Marks]****WCH14/01/Sample/Q15**

Consider synthesis of carboxylic acid 15 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q16. 16. Carboxylic Acid Synthesis & Reaction Step 16**[3 Marks]****WCH14/01/Sample/Q16**

Consider synthesis of carboxylic acid 16 (RCOOH) from nitrile RCN via acid hydrolysis.

- (a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

- (b) State the essential reaction condition for nitrile hydrolysis. [1]

Q17. 17. Carboxylic Acid Synthesis & Reaction Step 17**[3 Marks]****WCH14/01/Sample/Q17**

Consider synthesis of carboxylic acid 17 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q18. 18. Carboxylic Acid Synthesis & Reaction Step 18

[3 Marks]

WCH14/01/sample/Q18

Consider synthesis of carboxylic acid 18 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q19. 19. Carboxylic Acid Synthesis & Reaction Step 19

[3 Marks]

WCH14/01/sample/Q19

Consider synthesis of carboxylic acid 19 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q20. 20. Carboxylic Acid Synthesis & Reaction Step 20

[3 Marks]

WCH14/01/sample/Q20

Consider synthesis of carboxylic acid 20 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q21. 21. Carboxylic Acid Synthesis & Reaction Step 21

[3 Marks]

WCH14/01/sample/Q21

Consider synthesis of carboxylic acid 21 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q22. 22. Carboxylic Acid Synthesis & Reaction Step 22

[3 Marks]

WCH14/01/sample/Q22

Consider synthesis of carboxylic acid 22 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q23. 23. Carboxylic Acid Synthesis & Reaction Step 23

[3 Marks]

WCH14/01/sample/Q23

Consider synthesis of carboxylic acid 23 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q24. 24. Carboxylic Acid Synthesis & Reaction Step 24

[3 Marks]

WCH14/01/sample/Q24

Consider synthesis of carboxylic acid 24 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

Q25. 25. Carboxylic Acid Synthesis & Reaction Step 25

[3 Marks]

WCH14/01/sample/Q25

Consider synthesis of carboxylic acid 25 (RCOOH) from nitrile RCN via acid hydrolysis.

(a) Write the equation for the hydrolysis of RCN using dilute HCl . [2]

(b) State the essential reaction condition for nitrile hydrolysis. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Relative Acidity & Inductive Effects 26****[5 Marks]**
WCH14/01/Hard/Q26

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q27. 27. A* Challenge: Relative Acidity & Inductive Effects 27**[5 Marks]**
WCH14/01/Hard/Q27

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q28. 28. A* Challenge: Relative Acidity & Inductive Effects 28**[5 Marks]**
WCH14/01/Hard/Q28

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q29. 29. A* Challenge: Relative Acidity & Inductive Effects 29**[5 Marks]**
WCH14/01/Hard/Q29

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q30. 30. A* Challenge: Relative Acidity & Inductive Effects 30**[5 Marks]**
WCH14/01/Hard/Q30

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q31. 31. A* Challenge: Relative Acidity & Inductive Effects 31**[5 Marks]**
WCH14/01/Hard/Q31

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q32. 32. A* Challenge: Relative Acidity & Inductive Effects 32

[5 Marks]
WCH14/01/Hard/Q32

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q33. 33. A* Challenge: Relative Acidity & Inductive Effects 33

[5 Marks]
WCH14/01/Hard/Q33

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q34. 34. A* Challenge: Relative Acidity & Inductive Effects 34

[5 Marks]
WCH14/01/Hard/Q34

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q35. 35. A* Challenge: Relative Acidity & Inductive Effects 35

[5 Marks]
WCH14/01/Hard/Q35

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q36. 36. A* Challenge: Relative Acidity & Inductive Effects 36

[5 Marks]
WCH14/01/Hard/Q36

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q37. 37. A* Challenge: Relative Acidity & Inductive Effects 37

[5 Marks]
WCH14/01/Hard/Q37

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q38. 38. A* Challenge: Relative Acidity & Inductive Effects 38

[5 Marks]
WCH14/01/Hard/Q38

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q39. 39. A* Challenge: Relative Acidity & Inductive Effects 39

[5 Marks]
WCH14/01/Hard/Q39

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q40. 40. A* Challenge: Relative Acidity & Inductive Effects 40

[5 Marks]
WCH14/01/Hard/Q40

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q41. 41. A* Challenge: Relative Acidity & Inductive Effects 41

[5 Marks]
WCH14/01/Hard/Q41

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q42. 42. A* Challenge: Relative Acidity & Inductive Effects 42

[5 Marks]
WCH14/01/Hard/Q42

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q43. 43. A* Challenge: Relative Acidity & Inductive Effects 43

[5 Marks]
WCH14/01/Hard/Q43

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q44. 44. A* Challenge: Relative Acidity & Inductive Effects 44

[5 Marks]

WCH14/01/Hard/Q44

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q45. 45. A* Challenge: Relative Acidity & Inductive Effects 45

[5 Marks]

WCH14/01/Hard/Q45

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q46. 46. A* Challenge: Relative Acidity & Inductive Effects 46

[5 Marks]

WCH14/01/Hard/Q46

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q47. 47. A* Challenge: Relative Acidity & Inductive Effects 47

[5 Marks]
WCH14/01/Hard/Q47

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q48. 48. A* Challenge: Relative Acidity & Inductive Effects 48

[5 Marks]
WCH14/01/Hard/Q48

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q49. 49. A* Challenge: Relative Acidity & Inductive Effects 49**[5 Marks]**
WCH14/01/Hard/Q49

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

Q50. 50. A* Challenge: Relative Acidity & Inductive Effects 50**[5 Marks]**
WCH14/01/Hard/Q50

Compare the acid strength (K_a values) of ethanoic acid ($K_a = 1.74 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.0 \times 10^{-2}$).

(a) Explain the trend in K_a values in terms of the inductive effect of chlorine atoms. [3]

(b) Predict, with a reason, whether trichloroethanoic acid is a stronger or weaker acid than dichloroethanoic acid. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15C

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Carboxylic Acid Dimer Structure 1. (a) Two ethanoic acid molecules linked by 2 hydrogen bonds between C=O...H-O (2). 1. (b) Dimerisation effectively doubles the molecular size and strength of intermolecular forces (2).	[4]
Q2	Q2: 2. Reactions of Carboxylic Acids with NaHCO ₃ and PCl ₅ 2. (a) CH ₃ CH ₂ COOH + NaHCO ₃ → CH ₃ CH ₂ COONa + H ₂ O + CO ₂ (1). Effervescence / bubbles of gas (1). 2. (b) CH ₃ CH ₂ COOH + PCl ₅ → CH ₃ CH ₂ COCl + POCl ₃ + HCl (2). Misty / steamy white fumes of HCl (1).	[5]
Q3	Q3: 3. Carboxylic Acid Synthesis & Reaction Step 3 3. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 3. (b) Reflux with dilute acid / alkali (1).	[3]
Q4	Q4: 4. Carboxylic Acid Synthesis & Reaction Step 4 4. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 4. (b) Reflux with dilute acid / alkali (1).	[3]
Q5	Q5: 5. Carboxylic Acid Synthesis & Reaction Step 5 5. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 5. (b) Reflux with dilute acid / alkali (1).	[3]
Q6	Q6: 6. Carboxylic Acid Synthesis & Reaction Step 6 6. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 6. (b) Reflux with dilute acid / alkali (1).	[3]
Q7	Q7: 7. Carboxylic Acid Synthesis & Reaction Step 7 7. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 7. (b) Reflux with dilute acid / alkali (1).	[3]
Q8	Q8: 8. Carboxylic Acid Synthesis & Reaction Step 8 8. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 8. (b) Reflux with dilute acid / alkali (1).	[3]
Q9	Q9: 9. Carboxylic Acid Synthesis & Reaction Step 9 9. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 9. (b) Reflux with dilute acid / alkali (1).	[3]
Q10	Q10: 10. Carboxylic Acid Synthesis & Reaction Step 10 10. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 10. (b) Reflux with dilute acid / alkali (1).	[3]
Q11	Q11: 11. Carboxylic Acid Synthesis & Reaction Step 11 11. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 11. (b) Reflux with dilute acid / alkali (1).	[3]
Q12	Q12: 12. Carboxylic Acid Synthesis & Reaction Step 12 12. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 12. (b) Reflux with dilute acid / alkali (1).	[3]
Q13	Q13: 13. Carboxylic Acid Synthesis & Reaction Step 13 13. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 13. (b) Reflux with dilute acid / alkali (1).	[3]
Q14	Q14: 14. Carboxylic Acid Synthesis & Reaction Step 14 14. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 14. (b) Reflux with dilute acid / alkali (1).	[3]
Q15	Q15: 15. Carboxylic Acid Synthesis & Reaction Step 15 15. (a) RCN + HCl + 2H ₂ O → RCOOH + NH ₄ Cl (2). 15. (b) Reflux with dilute acid / alkali (1).	[3]

Q16	Q16: 16. Carboxylic Acid Synthesis & Reaction Step 16 16. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 16. (b) Reflux with dilute acid / alkali (1).	[3]
Q17	Q17: 17. Carboxylic Acid Synthesis & Reaction Step 17 17. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 17. (b) Reflux with dilute acid / alkali (1).	[3]
Q18	Q18: 18. Carboxylic Acid Synthesis & Reaction Step 18 18. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 18. (b) Reflux with dilute acid / alkali (1).	[3]
Q19	Q19: 19. Carboxylic Acid Synthesis & Reaction Step 19 19. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 19. (b) Reflux with dilute acid / alkali (1).	[3]
Q20	Q20: 20. Carboxylic Acid Synthesis & Reaction Step 20 20. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 20. (b) Reflux with dilute acid / alkali (1).	[3]
Q21	Q21: 21. Carboxylic Acid Synthesis & Reaction Step 21 21. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 21. (b) Reflux with dilute acid / alkali (1).	[3]
Q22	Q22: 22. Carboxylic Acid Synthesis & Reaction Step 22 22. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 22. (b) Reflux with dilute acid / alkali (1).	[3]
Q23	Q23: 23. Carboxylic Acid Synthesis & Reaction Step 23 23. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 23. (b) Reflux with dilute acid / alkali (1).	[3]
Q24	Q24: 24. Carboxylic Acid Synthesis & Reaction Step 24 24. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 24. (b) Reflux with dilute acid / alkali (1).	[3]
Q25	Q25: 25. Carboxylic Acid Synthesis & Reaction Step 25 25. (a) $\text{RCN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (2). 25. (b) Reflux with dilute acid / alkali (1).	[3]
Q26	Q26: 26. A* Challenge: Relative Acidity & Inductive Effects 26 26. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO^- anion (1). Stabilises the carboxylate anion, shifting $\text{RCOOH} \rightleftharpoons \text{RCOO}^- + \text{H}^+$ equilibrium to the right $\rightarrow$ higher K_a (1). 26. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl_3COO^- (1).	[5]
Q27	Q27: 27. A* Challenge: Relative Acidity & Inductive Effects 27 27. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO^- anion (1). Stabilises the carboxylate anion, shifting $\text{RCOOH} \rightleftharpoons \text{RCOO}^- + \text{H}^+$ equilibrium to the right $\rightarrow$ higher K_a (1). 27. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl_3COO^- (1).	[5]
Q28	Q28: 28. A* Challenge: Relative Acidity & Inductive Effects 28 28. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO^- anion (1). Stabilises the carboxylate anion, shifting $\text{RCOOH} \rightleftharpoons \text{RCOO}^- + \text{H}^+$ equilibrium to the right $\rightarrow$ higher K_a (1). 28. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl_3COO^- (1).	[5]

Q29	Q29: 29. A* Challenge: Relative Acidity & Inductive Effects 29 29. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 29. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q30	Q30: 30. A* Challenge: Relative Acidity & Inductive Effects 30 30. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 30. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q31	Q31: 31. A* Challenge: Relative Acidity & Inductive Effects 31 31. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 31. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q32	Q32: 32. A* Challenge: Relative Acidity & Inductive Effects 32 32. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 32. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q33	Q33: 33. A* Challenge: Relative Acidity & Inductive Effects 33 33. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 33. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q34	Q34: 34. A* Challenge: Relative Acidity & Inductive Effects 34 34. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 34. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q35	Q35: 35. A* Challenge: Relative Acidity & Inductive Effects 35 35. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 35. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q36	Q36: 36. A* Challenge: Relative Acidity & Inductive Effects 36 36. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 36. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]

Q37	Q37: 37. A* Challenge: Relative Acidity & Inductive Effects 37 37. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 37. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q38	Q38: 38. A* Challenge: Relative Acidity & Inductive Effects 38 38. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 38. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q39	Q39: 39. A* Challenge: Relative Acidity & Inductive Effects 39 39. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 39. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q40	Q40: 40. A* Challenge: Relative Acidity & Inductive Effects 40 40. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 40. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q41	Q41: 41. A* Challenge: Relative Acidity & Inductive Effects 41 41. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 41. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q42	Q42: 42. A* Challenge: Relative Acidity & Inductive Effects 42 42. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 42. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q43	Q43: 43. A* Challenge: Relative Acidity & Inductive Effects 43 43. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 43. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q44	Q44: 44. A* Challenge: Relative Acidity & Inductive Effects 44 44. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 44. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]

Q45	Q45: 45. A* Challenge: Relative Acidity & Inductive Effects 45 45. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 45. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q46	Q46: 46. A* Challenge: Relative Acidity & Inductive Effects 46 46. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 46. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q47	Q47: 47. A* Challenge: Relative Acidity & Inductive Effects 47 47. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 47. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q48	Q48: 48. A* Challenge: Relative Acidity & Inductive Effects 48 48. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 48. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q49	Q49: 49. A* Challenge: Relative Acidity & Inductive Effects 49 49. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 49. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]
Q50	Q50: 50. A* Challenge: Relative Acidity & Inductive Effects 50 50. (a) Chlorine is electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density from carboxylate C–O group, delocalising the negative charge on COO[–] anion (1). Stabilises the carboxylate anion, shifting RCOOH RCOO[–] + H⁺ equilibrium to the right → higher K_a (1). 50. (b) Stronger acid (higher K_a) (1). Three Cl atoms exert an even greater electron-withdrawing effect, further stabilising CCl₃COO[–] (1).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15C**FAQ #1: PCl_5 Test Observations****Category: Inorganic Reagents****COMMON EXAMINER TRAP / MISTAKE:**

Writing 'white precipitate' for PCl_5 test.

MODEL ANSWER / EXAMINER EXPECTATION:

PCl_5 reacts with $-\text{OH}$ groups to evolve HCl gas which forms MISTY / STEAMY WHITE FUMES when exposed to moist air. No precipitate is formed.

FAQ #2: NaHCO_3 vs Na_2CO_3 Reactions**Category: Carbonate Testing****COMMON EXAMINER TRAP / MISTAKE:**

Thinking NaHCO_3 gives different products than Na_2CO_3 .

MODEL ANSWER / EXAMINER EXPECTATION:

Both NaHCO_3 and Na_2CO_3 react with carboxylic acids to evolve CO_2 gas (effervescence). NaHCO_3 is preferred to test for carboxylic acids because weaker acids like phenol do NOT react with NaHCO_3 .

FAQ #3: Nitrile Hydrolysis Stoichiometry**Category: Nitrile Reactions****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting the water and acid stoichiometry in nitrile hydrolysis.

MODEL ANSWER / EXAMINER EXPECTATION:

Acidic hydrolysis: $\text{RCN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$ (requires 2 H_2O and 1 HCl). Alkaline hydrolysis: $\text{RCN} + \text{H}_2\text{O} + \text{NaOH} \rightarrow \text{RCOONa} + \text{NH}_3$.

FAQ #4: Inductive Effect on Carboxylic Acid Strength**Category: Acid Strength Factors****COMMON EXAMINER TRAP / MISTAKE:**

Claiming chlorine atoms make the O-H bond stronger.

MODEL ANSWER / EXAMINER EXPECTATION:

Chlorine is electron-withdrawing ($-\text{I}$ effect). It STABILISES THE CARBOXYLATE ANION (RCOO^-) by spreading out the negative charge. A more stable anion $\rightarrow$ equilibrium shifts right $\rightarrow$ STRONGER acid.

FAQ #5: Alkyl Group Effect on Acidity**Category: Inductive Effects****COMMON EXAMINER TRAP / MISTAKE:**

Expecting methanoic acid to be weaker than ethanoic acid.

MODEL ANSWER / EXAMINER EXPECTATION:

Alkyl groups (CH_3-) are electron-DONATING ($+\text{I}$ effect). They push electron density onto the COO^- group, destabilising the anion. Thus ethanoic acid is WEAKER than methanoic acid.

FAQ #6: Reduction of Carboxylic Acids**Category: Reduction Products****COMMON EXAMINER TRAP / MISTAKE:**

Using NaBH_4 to reduce carboxylic acids.

MODEL ANSWER / EXAMINER EXPECTATION:

NaBH₄ is TOO WEAK to reduce carboxylic acids! LiAlH₄ in dry ether MUST be used, reducing the carboxylic acid all the way to a PRIMARY ALCOHOL (RCH₂OH).

FAQ #7: Decarboxylation Reactions**Category: Decarboxylation****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting the reagent for removing a carboxyl group.

MODEL ANSWER / EXAMINER EXPECTATION:

Heating a carboxylic acid sodium salt (RCOONa) with soda lime (NaOH/CaO) removes CO₂ to form an alkane (RH) with ONE LESS CARBON atom.

FAQ #8: Carboxylic Acid Solubilities**Category: Hydrogen Bonding****COMMON EXAMINER TRAP / MISTAKE:**

Stating long-chain carboxylic acids are water-soluble.

MODEL ANSWER / EXAMINER EXPECTATION:

Carboxylic acids C₁-C₄ are miscible with water due to H-bonding. Beyond C₄, the hydrophobic non-polar alkyl chain dominates, drastically REDUCING water solubility.

FAQ #9: Esterification Equilibrium Shifting**Category: Reversible Reactions****COMMON EXAMINER TRAP / MISTAKE:**

Adding water to shift esterification equilibrium to the right.

MODEL ANSWER / EXAMINER EXPECTATION:

Esterification RCOOH + R'OH ⇌ RCOOR' + H₂O is reversible. To increase ester yield: remove water as it forms or use a large excess of alcohol/acid.

FAQ #10: Distinguishing Phenol from Carboxylic Acids**Category: Functional Group Tests****COMMON EXAMINER TRAP / MISTAKE:**

Assuming neutral FeCl₃ distinguishes carboxylic acids from phenols.

MODEL ANSWER / EXAMINER EXPECTATION:

Phenol reacts with neutral FeCl₃ to give a PURPLE solution, but does NOT react with NaHCO₃. Carboxylic acids react with NaHCO₃ to evolve CO₂ gas.